

Group Theory

Rising Action

1.

(a) The group of invertible $n \times n$ matrices over a finite field $\mathbb{F}_q$ is a finite group $\mathrm{GL}_n(\mathbb{F}_q)$. What is its order?

(b) What about the special linear group $\mathrm{SL}_n(\mathbb{F}_q)$; what is its order?

2. Let $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ be a short exact sequence of groups. Show that if A and C are solvable, then B is also solvable.

3. In this problem, fix an arbitrary finite nontrivial p -group G .

(a) Show that $Z(G) \neq \{e\}$.

(b) Show that G is nilpotent.

4. The additive group $\mathbb{Q}$ has the property that for any $q \in \mathbb{Q}$ and any $n \in \mathbb{Z}$, there exists an element $q' \in \mathbb{Q}$ such that $nq' = q$ (in particular $q' = q/n$). An abelian group with this property is called a *divisible* group.

(a) Show that any nontrivial divisible group is infinite.

(b) Show that if G is divisible, and N is a subgroup (normal since G is abelian), then G/N is divisible.

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5. (**January 2023 Problem 2**) We say a group G has *big finite quotients* if for every $k > 0$, there is a normal subgroup N of G such that G/N is finite and has order at least k .

(a) Give (with proof) an example of an infinite group that does *not* have big finite quotients.

(b) Show that $\mathrm{SL}_2(\mathbb{Z})$ (the group of 2×2 integer matrices with determinant 1) has big finite quotients.

6. (**January 2016 Problem 4**) Let D_k be the dihedral group of order $2k$ where $k \geq 3$.

(a) Show that the number of automorphisms of D_k is equal to $k \cdot \phi(k)$, where ϕ is Euler's ϕ -function.

(b) Let $\mathrm{Aut}(D_k)$ denote the group of automorphisms of D_k . What is the structure of $\mathrm{Aut}(D_k)$? Describe the group as explicitly as you can.

7. (August 2016 Problem 1) Let G be a group. By a maximal subgroup of G , we mean a subgroup $M \neq G$ such that the only subgroups containing M are M and G .

- (a) Describe all the maximal subgroups of the dihedral group of order $2p$ where p is an odd prime. How many are there?
- (b) Show that if a finite group G has only one maximal subgroup, then G is cyclic.
- (c) Show that if a maximal subgroup $M \triangleleft G$ is normal, then the index of M in G is finite and prime.

8. (January 2019 Problem 2) If G is a group and $x, y \in G$, then we let $\langle x, y \rangle$ denote the subgroup of G generated by x and y . For $x \in G$, define $\text{Cyc}_G(x) := \{y \in G \mid \langle x, y \rangle \text{ is cyclic}\}$.

- (a) Show by example that $\text{Cyc}_G(x)$ need not be a subgroup of G .
- (b) Let $\text{Cyc}(G) = \bigcap_{x \in G} \text{Cyc}_G(x)$. Show that $\text{Cyc}(G)$ is a subgroup of G .
- (c) Show that $\text{Cyc}(G)$ lies in the center of G .
- (d) Assume now that G is finite. Show that $\text{Cyc}(G)$ is cyclic.

9. (January 2023 Problem 5) Let p be a prime number and let G be the group of 2×2 matrices of the form

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix}, \quad a \in \mathbb{F}_p, \quad b \in \mathbb{F}_p^\times.$$

Fix a dimension n , and let $\rho: G \rightarrow \text{GL}_n(\mathbb{C})$ be a homomorphism. Put

$$x = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad y_b = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \quad (b \in \mathbb{F}_p^\times).$$

- (a) Show that $\rho(x)$ is diagonalizable, and that its eigenvalues are p th roots of unity.
- (b) Suppose $v \in \mathbb{C}^n$, $v \neq 0$ is an eigenvector of $\rho(x)$ of eigenvalue λ . Prove that for any $b \in \mathbb{F}_p^\times$, $\rho(y_b)v$ is also an eigenvector of $\rho(x)$, and find its eigenvalue.
- (c) Let v be as in part (b). Show that $W = \text{span}\langle \rho(y_b)v : b \in \mathbb{F}_p^\times \rangle$ is G -invariant (that is, $\rho(g)W \subseteq W$ for all $g \in G$), and that, if $\lambda \neq 1$, $\dim W = p - 1$.